

SkulHub

School Management System

System Documentation & User Flow Guide

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1. System Overview

SkulHub is a comprehensive, cloud-based school management system designed specifically for Kenyan schools and educational institutions. It provides 33+ integrated modules covering every aspect of school operations, from student admissions and fee collection to examination management and parent communication.

The system is built on Next.js 16 with TypeScript, uses Prisma ORM with PostgreSQL for data storage, and is hosted on Vercel's global CDN. It supports multi-tenancy, allowing unlimited schools to operate on a single platform with complete data isolation.

1.1 Key Features

- 33+ integrated modules covering all school operations
- Multi-tenancy with complete data isolation between schools
- 13 staff roles with granular, role-based access control
- Parent portal for fee checking, grades, and attendance
- Staff self-signup with principal approval workflow
- M-Pesa Daraja STK Push integration for fee collection
- CBC (Grade 1-8) and 8-4-4 (Form 1-4) curriculum support
- Printable invoices, receipts, and report cards with school letterhead
- Dark mode for eye comfort
- Auto-retry on connection errors with system health monitoring

1.2 Technology Stack

Component	Technology	Purpose
Framework	Next.js 16 (App Router)	Web application framework
Language	TypeScript 5	Type-safe development
Styling	Tailwind CSS 4 + shadcn/ui	Responsive UI design
Database	PostgreSQL (Neon)	Persistent data storage
ORM	Prisma 6	Database access layer
Hosting	Vercel	Global CDN deployment
Code Repository	GitHub (Private)	Version control
Authentication	HMAC-signed session tokens	Secure login
Payments	Safaricom Daraja API	M-Pesa STK Push
State Management	Zustand	Client-side state

2. User Roles & Access Control

SkulHub supports 13 distinct staff roles plus a platform super admin. Each role has specific module access permissions that can be further customized on a per-user basis by the school admin or principal.

2.1 Role Descriptions

Role	Key Responsibilities	Module Access Level
Super Admin	Platform owner, manages all schools	Super Admin dashboard, Settings
Admin	School administrator, full access	All 33+ modules
Principal	Head of school, full access	All modules except system settings
Deputy Principal	Assists principal, academic focus	Academic + administration modules
Bursar	Fee collection, finance	Finance, procurement, payroll
Teacher	Teaching, attendance, grades	Students, academics, attendance, exams
Librarian	Library management	Library, inventory requests
Nurse	Health records	Health module, students
Matron	Boarding supervision	Hostel, health, discipline
Secretary	Administration, communications	Admissions, students, communications
Admissions Clerk	Student admissions	Admissions, students, ID cards
Bus Driver	Transport management	Transport module
Gate Man	Security, visitor tracking	Visitors, transport
Cook	Kitchen, meal management	Cafeteria, inventory requests

3. User Flows

This section describes the step-by-step workflow for each major user type, from login to completing their primary tasks.

3.1 School Admin Flow

Login Process

- Visit skulhub.co.ke and click 'Sign In'
- Enter school code (e.g., SKH-2024-001) and click 'Continue'
- Enter admin email and password
- System authenticates and redirects to the dashboard

Daily Tasks

- View dashboard for school-wide statistics (students, fees, attendance)
- Manage staff approvals (approve/reject new staff signups)
- Configure module access for individual users (Settings > Module Access)
- Set up fee structures and generate bulk invoices
- Monitor financial reports and collection rates
- Manage school settings (general, academic, notifications, M-Pesa)

3.2 Principal Flow

Login Process

- Same as admin: school code + email + password

Daily Tasks

- Review dashboard for key metrics and trends
- Approve/reject staff registration requests
- Monitor attendance trends and student performance
- Publish announcements for staff, students, and parents
- Review discipline incidents and take action
- Oversee hostel/boarding allocations
- Generate and review comprehensive reports

3.3 Bursar Flow

Login Process

- Same as admin: school code + email + password

Daily Tasks

- View financial dashboard: total billed, collected, outstanding
- Generate invoices for students (individual or bulk)

- Record payments (M-Pesa, Cash, Bank Transfer, Cheque)
- Cash payments get auto-generated reference numbers (CSH-YYYYMMDD-XXXX)
- Print invoices and payment receipts with school letterhead
- Initiate M-Pesa STK Push for fee collection (if Daraja configured)
- Manage expenses and scholarships
- Generate financial reports for the principal

3.4 Teacher Flow

Login Process

- Same as admin: school code + email + password

Daily Tasks

- Mark daily attendance for assigned class
- Record grades for assessments and exams
- Add external exam marks (KCSE, KCPE, Mock) for students
- Change wrongly assigned subjects on assessments
- Create and assign homework
- Create lesson plans and schemes of work
- View timetable for teaching schedule
- Submit inventory requests for teaching materials

3.5 Parent Flow

Login Process

- Visit skulhub.co.ke and click 'Parent? Access parent portal'
- Enter school code, student admission number, and guardian phone number
- No password required - verified by matching phone number

Portal Features

- View child's fee summary (billed, paid, balance)
- Check recent attendance records
- View recent grades and exam results
- Read school announcements
- Check upcoming events

3.6 Staff Self-Signup Flow

Registration Process

- Staff member visits skulhub.co.ke and clicks 'Staff? Sign up to join your school'
- Enters the school code (provided by the school)
- Fills personal details: name, email, password, phone, role, gender, qualification, specialization
- Account is created with 'Pending' status
- Principal/Admin sees the request in 'Staff Approvals' module
- Principal approves or rejects (with reason) the request
- If approved: staff can log in with their credentials
- If rejected: staff sees the rejection reason when trying to log in

3.7 New School Registration Flow

Registration Process

- School visits skulhub.co.ke and clicks 'Start Free Trial'
- Completes 6-step registration wizard:
 - Step 1: School basics (name, category, email, phone)
 - Step 2: School type (level, KNEC code, year established)
 - Step 3: Gender & location (Boys/Girls/Mixed, county, address)
 - Step 4: School identity (motto, brand color, logo)
 - Step 5: Principal account (name, email, password)
 - Step 6: Review & submit
- System auto-generates school code (e.g., SKH-2026-003)
- School gets 30-day free trial
- Principal can log in immediately and start using the system

3.8 Super Admin (Platform Owner) Flow

Login Process

- On the login screen, press Ctrl+Shift+A
- Enter access code: skulhub-super-2026
- Enter super admin email and password

Platform Management Tasks

- View all registered schools with stats (students, staff, revenue)

- Activate, suspend, upgrade, or delete schools
- Monitor platform-wide revenue and growth
- View and manage demo requests from the website
- Track referrals and commission payouts
- Monitor system health (database, storage, network, auth, M-Pesa)

4. Module Overview

SkulHub includes 33+ modules organized into functional groups. Each module is accessible from the sidebar based on the user's role and permissions.

4.1 Module List

Module	Group	Description
Dashboard	Overview	Real-time school stats, charts, and activity feed
Admissions	People	Manage student applications and enrollment pipeline
Students	People	Student profiles, enrollment, guardian information
Staff & Teachers	People	Staff management, assignments, departments
Staff Approvals	People	Approve/reject staff self-signup requests
Alumni	People	Graduate tracking and donations
Academics	Academic	Classes, subjects, streams, departments
Attendance	Academic	Daily marking, reports, trends
Examinations	Academic	Question banks, CATs, grading, external marks
Report Cards	Academic	KCSE-style printable report cards
Lesson Plans	Academic	Schemes of work and lesson planning
Homework	Academic	Assignment creation and tracking
Timetable	Academic	Weekly class schedule with visual grid
Finance & Fees	Administration	Invoices, payments, M-Pesa, expenses
Payroll	Administration	Staff salaries and payslips
Communications	Administration	Announcements, SMS/email queue
Library	Administration	Books, borrowing, returns
Transport	Administration	Routes, vehicles, assignments
Inventory	Administration	Assets, stocktake, restock, purchase orders
Health & Wellness	Academic	Medical records, clinic visits
Hostel & Boarding	Academic	Dormitories, beds, allocations

Module	Group	Description
Events & Activities	Academic	Calendar and event management
Discipline	Academic	Incidents and sanctions tracking
Cafeteria & Meals	Administration	Menu and dining attendance
Procurement	Administration	Suppliers and purchase orders
Facilities	Administration	Facility booking management
Visitors & Gate	Administration	Security and visitor tracking
Staff Room Board	Insights	Live display screen for staff
Appraisals	Insights	Staff performance reviews
Feedback	Insights	Surveys and ratings
ID Cards	Insights	Printable student/staff IDs
Data Import	Insights	Bulk data migration
Reports	Insights	Analytics and reporting
Settings	Insights	System configuration

5. Security & Data Protection

5.1 Authentication

- HMAC-signed session tokens (7-day expiry)
- Passwords hashed with bcrypt (10 rounds)
- HttpOnly cookies prevent XSS token theft
- Per-user module access overrides (admin can tick/untick modules per user)

5.2 Data Isolation

- Each school has a unique schoolId
- All queries are scoped to the user's schoolId
- Schools cannot access each other's data
- Super admin can see aggregate data across all schools

5.3 Database Security

- PostgreSQL hosted on Neon (encrypted at rest)
- SSL/TLS connection (sslmode=require)
- Automated daily backups (Neon)
- Point-in-time recovery available

6. Deployment Architecture

6.1 Infrastructure

Component	Provider	Location	Purpose
Code Repository	GitHub	Cloud	Version control (private repo)
Web Hosting	Vercel	Global CDN	Next.js deployment
Database	Neon	Singapore (ap-southeast-1)	PostgreSQL
DNS	Cloudflare	Global	Domain name resolution
Domain	Truehost	Kenya	skulhub.co.ke registration

6.2 Deployment Flow

- Developer pushes code to GitHub
- Vercel auto-detects the push and starts building
- Vercel runs: prisma generate -> next build
- If build succeeds, deployment goes live automatically
- Changes are visible on skulhub.co.ke within 2-3 minutes

7. Demo Credentials

The following credentials are for the demo school (SkulHub Academy, code: SKH-2024-001):

Role	Email	Password
Admin	admin@skulhub.ac.ke	admin123
Principal	principal@skulhub.ac.ke	principal123
Deputy Principal	deputy@skulhub.ac.ke	deputy123
Bursar	bursar@skulhub.ac.ke	bursar123
Teacher	teacher@skulhub.ac.ke	teacher123
Librarian	librarian@skulhub.ac.ke	librarian123
Nurse	nurse@skulhub.ac.ke	nurse123
Admissions	admissions@skulhub.ac.ke	admissions123
Matron	matron@skulhub.ac.ke	matron123
Secretary	secretary@skulhub.ac.ke	secretary123
Bus Driver	driver@skulhub.ac.ke	driver123

Role	Email	Password
Gate Officer	gate@skulhub.ac.ke	gate123
Cook	cook@skulhub.ac.ke	cook123
Super Admin	superadmin@skulhub.ac.ke	superadmin123

Super Admin access: Press Ctrl+Shift+A on the login screen, then enter access code: skulhub-super-2026

8. System Sign-off

By signing below, the parties acknowledge that they have reviewed the SkulHub School Management System documentation, understand the system's capabilities, user flows, and security measures, and agree to the system being deployed and used as described above.

8.1 Acknowledgement

- The system has been reviewed and meets the school's requirements.
- All user roles and access permissions are understood and agreed upon.
- The school agrees to maintain the confidentiality of all login credentials.
- The school understands that data is stored securely on cloud servers (Vercel + Neon PostgreSQL).
- The school agrees to the pricing plan selected below.
- The system is accepted for production use.

8.2 Subscription Details

Field	Details
School Name	_____
School Code	_____
Selected Plan	Starter (KES 2,500/mo) / Standard (KES 5,000/mo) / Premium (KES 10,000/mo)
Billing Cycle	Monthly / Annually (2 months free)
Trial End Date	_____
Subscription Start Date	_____

8.3 Authorised Signatures

School Representative	SkulHub Platform Owner
Name: _____	Name: _____
Title: _____	Title: _____
Signature: _____	Signature: _____
Date: _____	Date: _____
Stamp:	Stamp:

8.4 Terms & Conditions

- The school agrees to pay the monthly subscription fee by the 5th of each month.
- The subscription can be cancelled at any time with 30 days' notice.
- Data export is available on request at no additional cost.
- SkulHub is not liable for data loss due to school negligence (e.g., sharing credentials).
- Support is available via email (info@skulhub.co.ke) and phone (0742 340 924).
- Free training is provided for the first session (up to 20 staff).
- The school is responsible for maintaining accurate data in the system.
- Any custom feature requests will be quoted separately.

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